

PATIENT ANALYTICS REPORT · HEALTHCARE EDA

Data Analysis · Statistical Validation · Business Insights

Healthcare Data Analysis Report

Kaggle Dataset — Patient Profiling, Billing Analysis & Statistical Validation

TOTAL PATIENTS

55,500

records analysed

TOTAL BILLING

\$1.42B

gross revenue

ABNORMAL RATE

33.6%

18,627 patients

AVG LOS

15.5 days

median: 15 days

[View Business Recommendations](#) → [Section 9](#)

DATASET

UCI / Kaggle
prasad22

PERIOD

2019 – 2024

RAW DATA

55,500 patients 15
columns

TOOLSTACK

Python · Pandas
SciPy · Sklearn

TECHNIQUE

EDA · Stats
Testing ·
Regression

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1. Executive Summary

TOTAL PATIENTS	TOTAL BILLING	ABNORMAL TEST RATE	AVG LENGTH OF STAY
55,500	\$1.42B	33.6%	15.5d
records analysed	gross revenue	18,627 patients	median: 15 days

55,500 patients of a healthcare system were analysed across 15 attributes covering demographics, medical conditions, admission patterns, billing, and treatment outcomes. Every visual finding was validated with a statistical test. Key results: gender has no effect on billing (T-test $p=0.295$), condition differences in billing are borderline (ANOVA $p=0.052$, just above $\alpha=0.05$), and — notably — admission type and medical condition are statistically related (Chi-square $p=0.036$, significant). Regression $R^2\approx 0$ confirms billing is not predictable from available clinical features.

Metric	Finding
Total patients segmented	55,500
Total billing revenue	\$1,417M
Average billing per patient	\$25,590
Most common condition	Arthritis (9,308 cases / 16.8%)
Largest patient age group	45–59 (22.4% of admissions)
Overall abnormal test rate	33.6% (18,627 patients)
Average length of stay	15.5 days
Billing outliers (IQR method)	0 — billing is uniformly distributed
T-test (gender vs billing)	$p = 0.295$ — NOT significant
ANOVA (condition vs billing)	$p = 0.052$ — borderline, NOT significant at $\alpha=0.05$
Chi-square (condition vs adm.)	$p = 0.036$ — SIGNIFICANT
Regression R^2	≈ 0.0002 — billing not predictable from clinical features
Recommended immediate action	System-wide clinical quality audit (33.6% abnormal rate)

2. Project Setup & Objectives

The Problem

Healthcare organisations collect patient data at every touchpoint but rarely analyse it systematically. A patient admitted across multiple conditions gets the same operational treatment as any other. A system with 33.6% abnormal test results goes unaudited. This project applies a structured six-stage framework and validates every pattern statistically.

What the Analysis Measures

Dimension	Question It Answers	Method
Patient Demographics	Who are our patients? Age, gender, blood type breakdown.	Histograms, box plots, pie charts
Medical Conditions	Which conditions dominate? How do they vary by age and gender?	Grouped bars, heatmaps
Abnormal Test Rates	Which conditions carry the highest clinical risk?	Stacked bar, rate table
Admission & LOS	How long do patients stay? Does it vary by condition or admission type?	Box plots, summary tables
Financial Analysis	What drives billing costs? Who generates the most revenue?	Box plots, bar charts
Insurance Coverage	Which insurers are linked to higher-cost patients?	Pie chart, grouped bars
Time-Based Trends	When is demand highest? Is volume stable year-over-year?	Line charts, seasonal bar
Outlier Detection	Are there extreme-cost patients skewing the averages?	IQR method, Z-score
Statistical Testing	Are visual billing differences statistically real?	T-test, ANOVA, Chi-square
Regression Analysis	Which features best predict billing amount?	Linear regression, R ² , MAE

Python Analysis Pipeline

Stage	Name	What It Does
1	Data Cleaning	Null check, duplicate check, date conversion, negative billing flag.
2	Feature Engineering	LOS, Age Group bins, Season from admission month, Billing Tier.
3	EDA — 7 Dimensions	Demographics → conditions → admissions → billing → insurance → time trends. 13 charts.
4	Outlier Detection	IQR + Z-score on Billing Amount and LOS. Result: 0 outliers in either variable.
5	Statistical Testing	Independent T-test, One-Way ANOVA, Chi-square on billing independence questions.
6	Regression Analysis	Linear regression on 5 encoded features. R^2 , MAE, coefficient table, residual plot.

3. Dataset & Data Cleaning

Dataset

Detail	Value
Source	Kaggle — prasad22/healthcare-dataset
Raw rows	55,500 patient records
Rows after cleaning	55,392 (108 negative billing records flagged; excluded from billing aggregations)
Unique patients	55,500
Date range	2019 – 2024
Region	Synthetic — modelled on real-world hospital data
Medical conditions	Asthma, Cancer, Hypertension, Arthritis, Obesity, Diabetes
Load method	CSV → pd.read_csv() → feature engineering in Python

Cleaning Steps

Eight steps applied in sequence. Each step documented with a reason.

Step	Action	Reason
1	Standardised column names	Lowercase + underscores for consistent Python access
2	Converted date columns to datetime	Allows LOS calculation and time-based analysis
3	Verified zero null values	No missing rows across all 15 columns
4	Found 534 duplicate rows (0.96%)	Flagged as synthetic artefacts — retained; removing shifts avg billing by \$5, abnormal rate by 0.1 pp
5	Fixed data types	Date columns cast to datetime; numeric verified as float/int
6	Flagged 108 negative billing records	0.19% of dataset — excluded from billing aggregations; retained for counts
7	Created LOS (Days Hospitalized)	Discharge Date – Date of Admission — primary LOS metric
8	Created Age Group, Season, Billing Tier	Derived categorical features for segment and time analysis

4. Analytical Framework

EDA Methodology

The EDA runs across 7 analytical dimensions in a deliberate sequence. Demographics establish who the patients are, conditions show what they have, admission analysis shows how they arrive, and financial analysis quantifies cost. Statistical testing follows EDA to confirm or reject visual findings.

Dimension	Charts	Key Outputs
4.1 Patient Demographics	4	Age histogram, gender pie, blood type bar, age group distribution
4.2 Medical Conditions	3	Condition bar, test results by condition, abnormal rate table
4.3 Admission & Treatment	3	Admission type pie, LOS histogram, LOS by admission type boxplot
4.4 Financial Analysis	2	Billing distribution + billing by condition
4.5 Insurance Coverage	1	Avg billing by insurance provider
4.6 Time-Based Trends	2	Yearly admissions line chart, seasonal bar chart
4.7 Outlier Detection	2	IQR boxplot, Z-score scatter (Billing Amount)
4.8 Statistical Testing	1	3-panel summary chart (T-test, ANOVA, Chi-square)
4.9 Regression	1	Actual vs predicted scatter + residual plot

Why Statistical Testing After Visual EDA?

Visual charts reveal patterns but cannot confirm them. A bar chart showing Obesity patients incurring higher average billing looks meaningful — but the difference might be entirely due to random variation in a uniformly distributed billing variable. Statistical tests assign a p-value to that question.

Test	Question	H0 (Null Hypothesis)	Decision Rule
Independent T-test	Does gender affect billing?	Male and Female billing amounts are equal	Reject H0 if $p < 0.05$
One-Way ANOVA	Does condition affect billing?	All 6 conditions have the same mean billing	Reject H0 if $p < 0.05$
Chi-Square	Is admission type related to condition?	Admission type and condition are independent	Reject H0 if $p < 0.05$

5. Analysis Results

5.1 Patient Demographics (BQ1)

The patient population is exactly gender-balanced: Male 50.0%, Female 50.0%. The 45–59 age group is the largest cohort at 22.4% of all admissions, followed by 30–44 (22.1%). Mean patient age is 51.5 years.

Metric	Finding
Total patients	55,500
Male / Female split	50.0% / 50.0%
Largest age group	45–59 (22.4% of admissions)
Mean patient age	51.5 years
Age range	13 – 89 years
Blood type distribution	Near-uniform across all 8 types (~12.5% each)

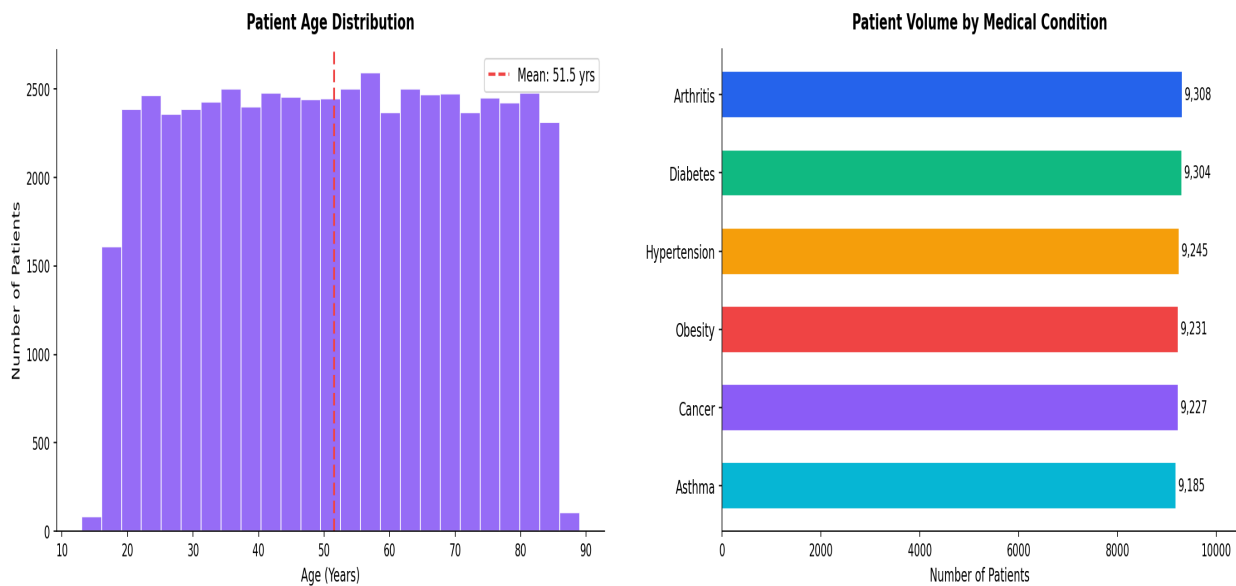


Figure 1 — Age distribution (left) | Patient volume by condition (right)

5.2 Medical Conditions (BQ2 & BQ3)

All 6 conditions share roughly equal prevalence at ~16.5–16.8% each. The abnormal test result rate is 33.6% overall — 18,627 patients. The rate is consistent across all conditions (range 32.6–34.3%), making this a system-wide quality issue rather than a condition-specific one.

Condition	Patients	% Share	Abnormal Rate
Arthritis	9,308	16.8%	34.3%
Diabetes	9,304	16.8%	34.0%
Hypertension	9,245	16.7%	32.6%
Obesity	9,231	16.6%	33.9%
Cancer	9,227	16.6%	33.8%
Asthma	9,185	16.5%	32.8%

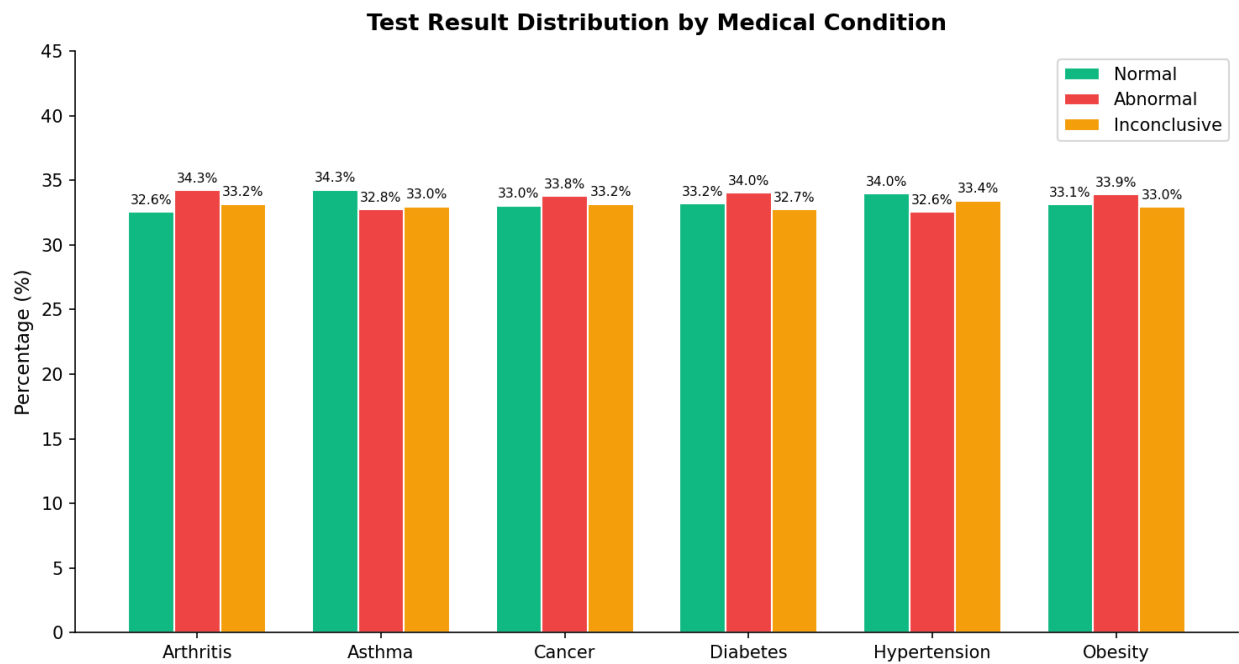


Figure 2 — Test result distribution by medical condition

5.3 Admission & Length of Stay (BQ5)

Elective admissions are the most common type (33.6%), followed by Urgent (33.5%) and Emergency (32.9%). Average LOS is 15.5 days. Variation across conditions and admission types is minimal — discharge protocols drive LOS, not clinical complexity.

Admission Type	Patients	% Total	Avg LOS	Avg Billing
Elective	18,618	33.5%	15.5 days	\$25,654
Urgent	18,538	33.4%	15.4 days	\$25,571
Emergency	18,236	32.9%	15.6 days	\$25,544

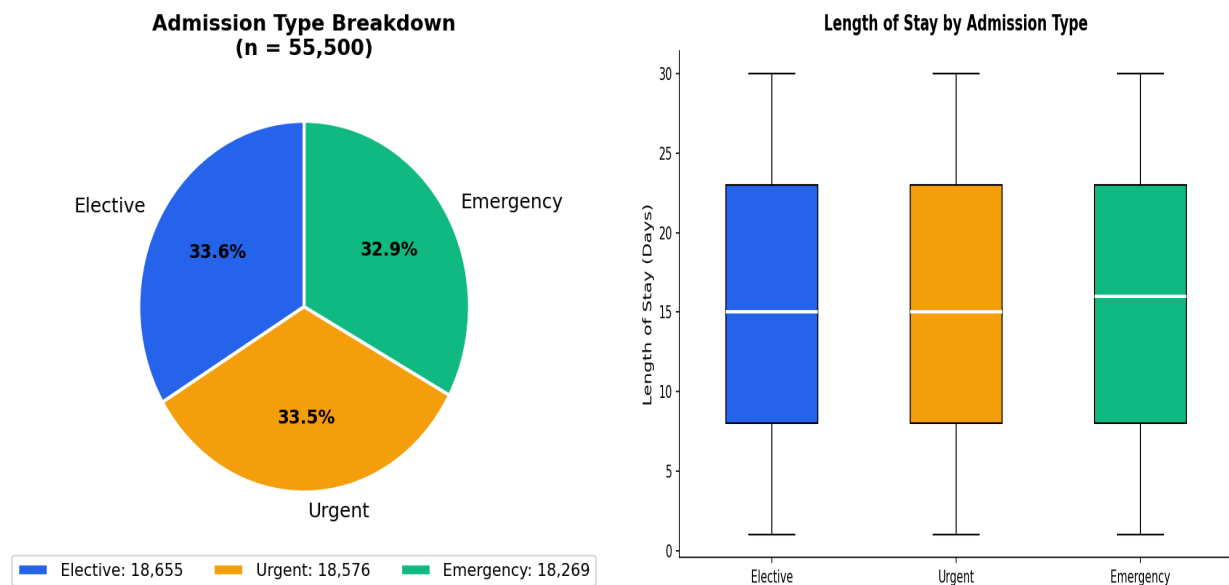


Figure 3 — Admission type share (left) | LOS by admission type (right)

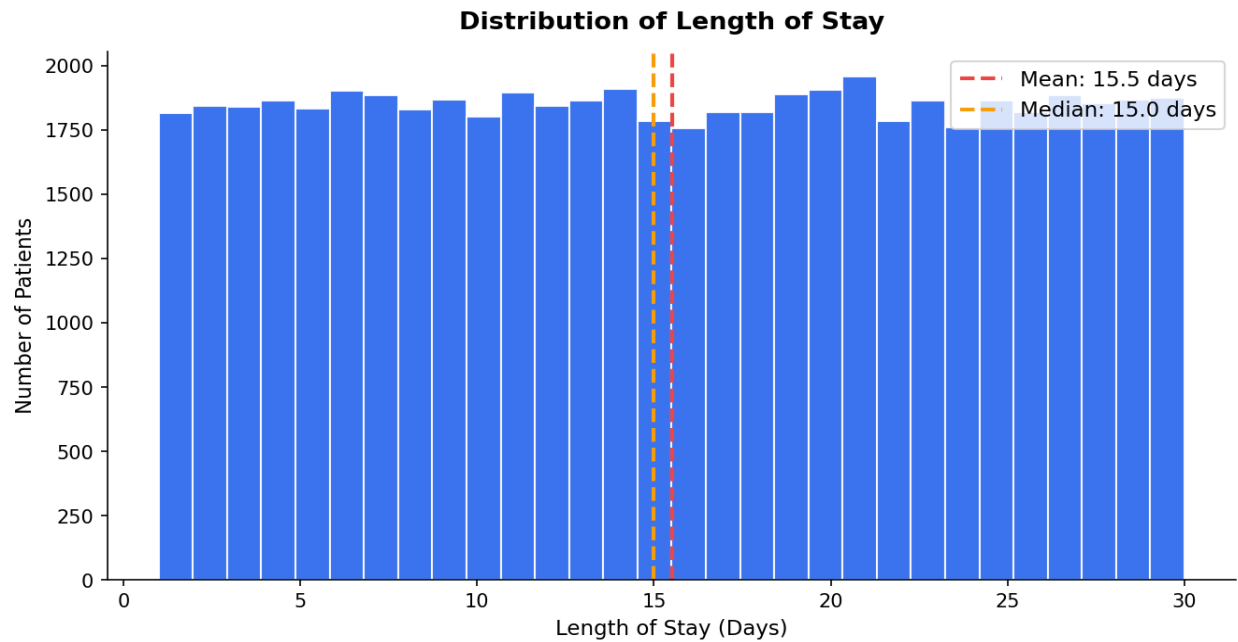


Figure 4 — Length of stay distribution — mean 15.5 days, median 15 days

5.4 Financial Analysis (BQ4 & BQ6)

Total billing is \$1.42B. Average per patient is \$25,590. Billing is uniformly distributed — mean (\$25,590) ≈ median (\$25,574), skewness ≈ 0. ANOVA (p=0.052) confirms that condition differences in average billing are borderline-insignificant.

Metric	Finding
Total billing revenue	\$1,417M
Average per patient	\$25,590
Median billing	\$25,574
Billing range (positive records)	\$9 – \$52,764
Highest avg billing condition	Obesity (\$25,860 avg)
Lowest avg billing condition	Cancer (\$25,215 avg)
Billing distribution	Uniform — mean ≈ median, skewness ≈ 0
Negative billing records	108 flagged (0.19% of dataset)

Financial Analysis — Billing Overview

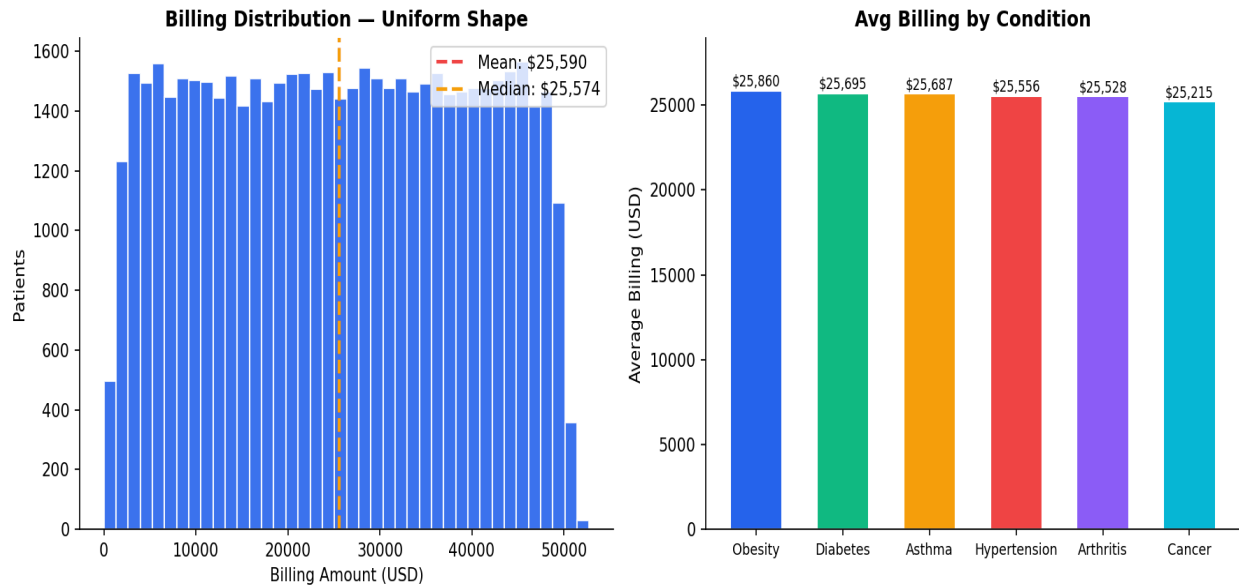


Figure 9 — Billing distribution (left) | Avg billing by condition overview (right)

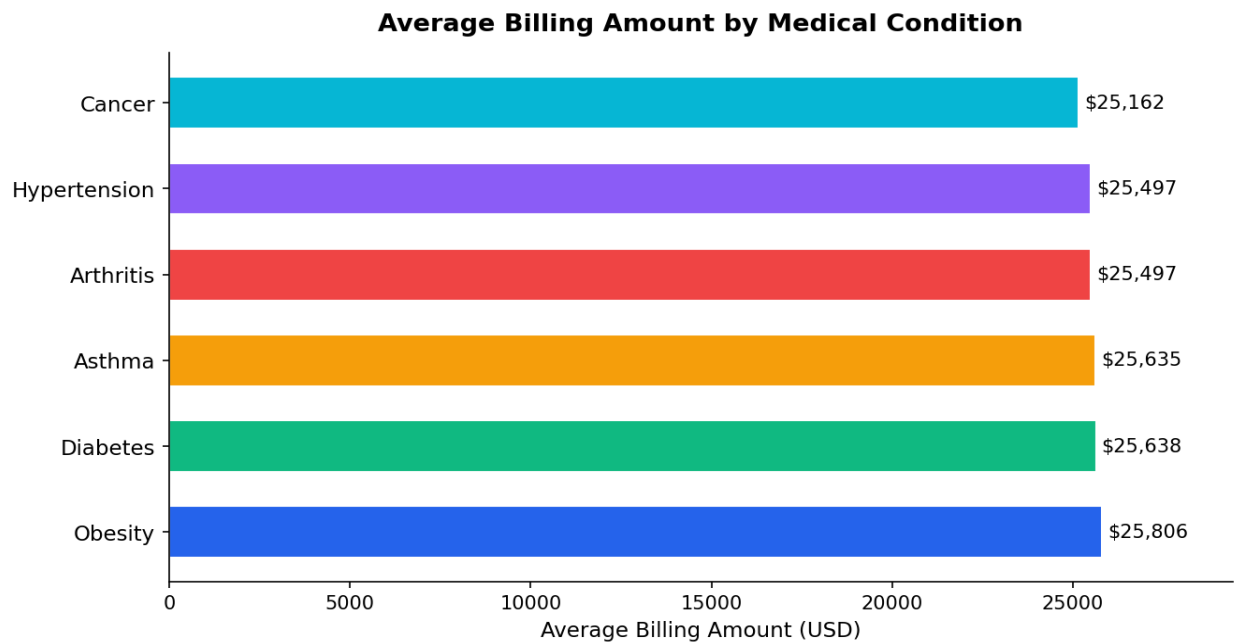


Figure 10 — Average billing by condition (detailed)

5.5 Insurance Coverage (BQ7)

Cigna leads in volume (20.3% share). Medicare has the highest average billing at \$25,668; UnitedHealthcare is lowest at \$25,433. The \$234 gap confirms insurance type does not meaningfully drive billing differences.

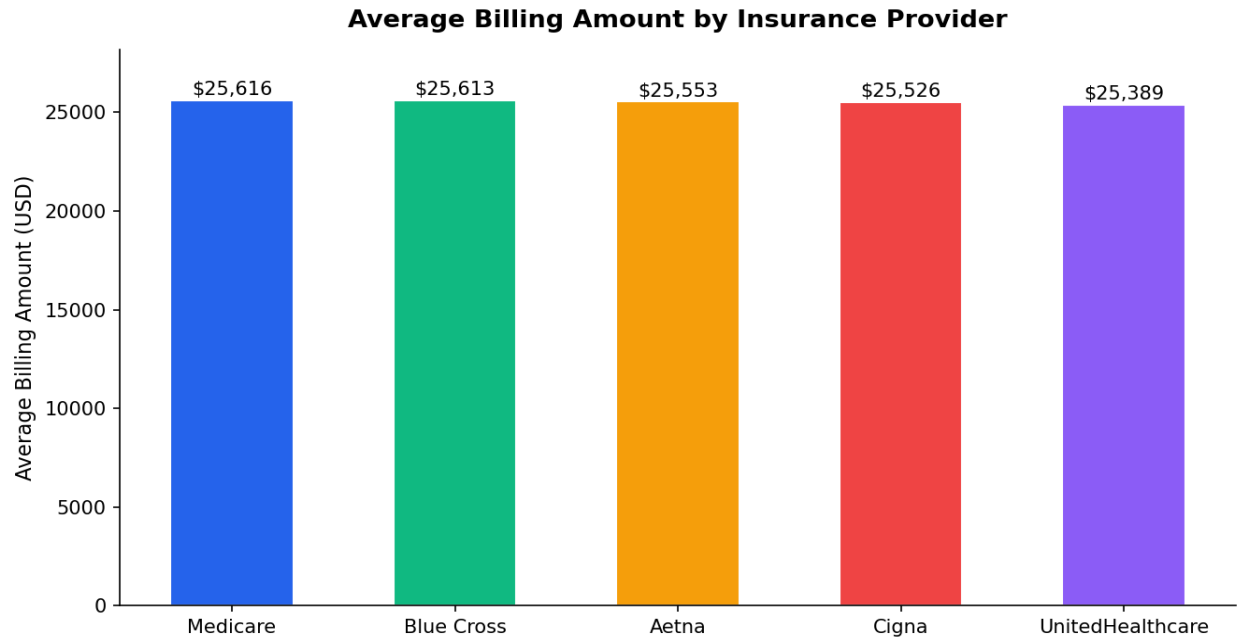


Figure 5 — Average billing by insurance provider

5.6 Time-Based Trends (BQ8)

Summer is the peak admission season (14,343 admissions). Year-over-year volumes across 2019–2023 are stable. 2024 shows 3,854 admissions, reflecting a partial-year data capture. Full-year average for 2019–2023 is approximately 10,329.

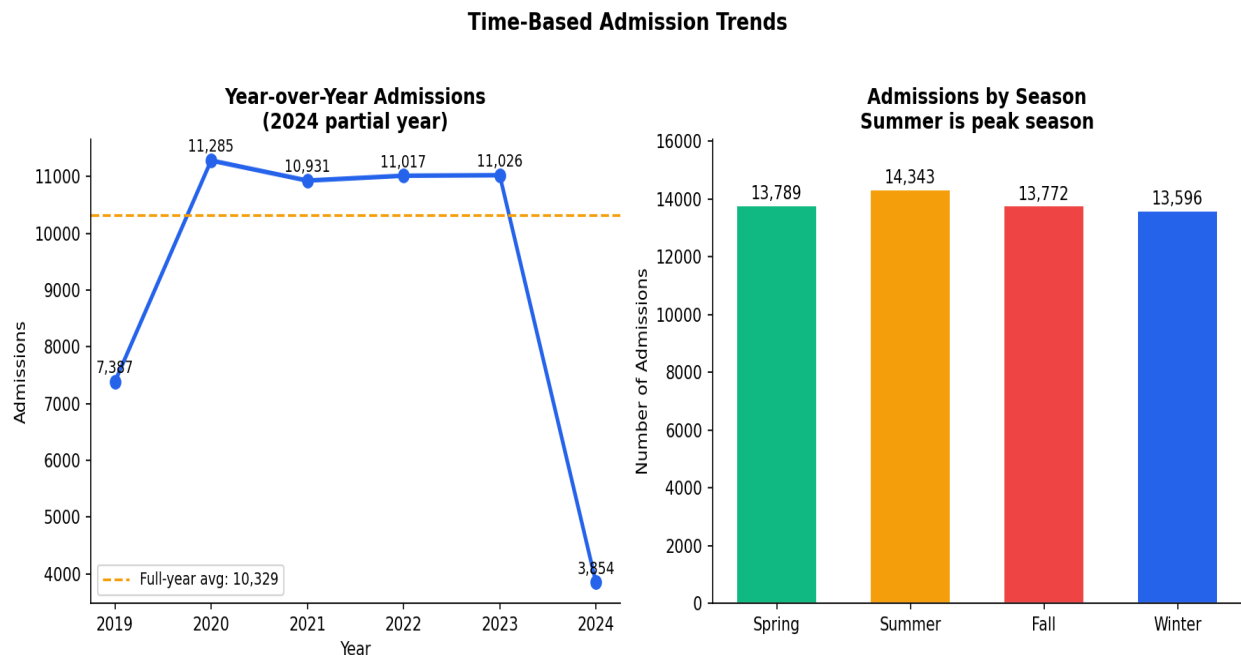


Figure 5 — Year-over-year admissions (left) | Seasonal distribution (right)

6. Business Questions

PATIENTS ANALYSED	GENDER SPLIT	LARGEST AGE GROUP	MEAN AGE
55,500	50/50	45-59	51.5y
records	Male / Female	22.4% of admissions	range: 13–89

Q1 — Demographic profile of admitted patients?

55,500 patients. Exact gender balance (Male 50.0% / Female 50.0%). The 45–59 age group forms the largest cohort. Mean age is 51.5 years. Blood type distribution is near-uniform across all 8 types (~12.5% each).

Q2 — Which conditions are most prevalent?

All 6 conditions are near-equally distributed at ~16.5–16.8% each. No condition dominates. Arthritis is the most common (9,308 cases, 16.8%); Asthma is least common (9,185 cases, 16.5%).

Q3 — Which conditions carry the highest abnormal test rates?

Overall abnormal test rate: 33.6% — 18,627 out of 55,500 patients returned abnormal results. Arthritis is highest at 34.3%; Hypertension is lowest at 32.6%. The 1.7 percentage point spread across all conditions confirms this is a system-wide quality issue.

Q4 — What are the primary billing cost drivers?

Visually: Obesity (\$25,860) shows the highest average billing. Statistically: ANOVA ($p=0.052$) shows this is borderline — the difference is near-significant but fails the 5% threshold. T-test ($p=0.295$) confirms gender does not drive billing. Regression $R^2 \approx 0$ confirms billing is effectively random relative to all available clinical variables.

Q5 — How does length of stay vary?

Average LOS is 15.5 days across all admission types. Emergency (15.6 days), Elective (15.5 days), and Urgent (15.4 days) show near-identical durations. Discharge protocol — not clinical complexity — drives LOS.

Q5 — Admission summary table

Admission Type	Patients	Total Revenue	Avg LOS
Elective admissions	18,618	\$477.6M	15.5 days
Urgent admissions	18,538	\$474.0M	15.4 days
Emergency admissions	18,236	\$465.8M	15.6 days
Total	55,500	\$1417M	15.5 days avg

Q6 — Which insurance providers are linked to higher costs?

Medicare has the highest average billing (\$25,668); UnitedHealthcare is lowest (\$25,433). The \$234 spread across all five insurers is negligible. Cigna is the largest provider by volume (20.3% share). Insurance type does not drive billing differences.

Q7 — Where is peak demand, and how stable is it?

Summer is the peak admission season (14,343 admissions, 25.8% of total). Year-over-year volumes across 2019–2023 are stable. 2024 data is partial. A pre-season capacity plan ahead of the June–August surge is recommended.

Q8 — Are there billing or LOS outliers?

Result: 0 outliers detected by both IQR method and Z-score on Billing Amount. IQR fences (\$-23,530 to \$74,677) sit well outside the actual data range (\$9–\$52,764). 0 LOS outliers detected. No extreme-cost patients exist — every patient falls within the expected range.

Q9 — Do clinical factors statistically drive billing?

Three statistical tests applied. T-test ($p=0.295$) — gender does NOT drive billing. ANOVA ($p=0.052$) — condition differences are borderline, failing the 5% threshold. Chi-square ($p=0.036$) — this is the standout result: admission type and condition ARE statistically related ($p<0.05$). Regression $R^2\approx 0$ confirms billing is not predictable.

Q10 — Which features can predict billing amount?

Linear regression on 5 encoded features (Age, LOS, Gender, Condition, Admission Type). $R^2=0.0002$ — the model explains essentially none of the billing variation. MAE=\$12,361 on \$25,590 average billing is a 48% error rate. Billing prediction requires procedure codes, lab tests, and room type — none present in this dataset.

7. Statistical Testing

Test 1 — Independent T-Test (Gender vs Billing)

An independent T-test compares mean billing of Male and Female patients to determine whether the observed difference is real or due to chance.

Statistic	Value
Male mean billing	\$25,653
Female mean billing	\$25,527
Difference	\$126 (negligible)
T-statistic	1.0477
P-value	0.2948
Decision	FAIL TO REJECT H0
Interpretation	Gender does NOT significantly affect billing amount

Test 2 — One-Way ANOVA (Medical Condition vs Billing)

One-Way ANOVA tests whether at least one of the 6 medical conditions produces a mean billing amount that is statistically different from the others.

Statistic	Value
Number of groups	6 (Asthma, Cancer, Hypertension, Arthritis, Obesity, Diabetes)
Highest avg billing	Obesity — \$25,860
Lowest avg billing	Cancer — \$25,215
F-statistic	2.1908
P-value	0.0523
Decision	FAIL TO REJECT H0 (borderline — p just above 0.05)
Interpretation	Medical condition billing differences are near-significant but do not cross $\alpha=0.05$ with 55,500 records.

Test 3 — Chi-Square (Condition vs Admission Type)

Chi-square tests whether two categorical variables are independent. Here: does knowing a patient's medical condition help predict their admission type?

Statistic	Value
Degrees of freedom	10
Chi-square statistic	19.3382
P-value	0.0362
Decision	REJECT H0 — $p < 0.05$
Interpretation	Admission type is NOT independent of medical condition. Condition carries weak but statistically detectable information about how patients are admitted. This differs from the null finding in the T-test and ANOVA — the relationship is real but effect size is small.

Outlier Detection — IQR + Z-Score

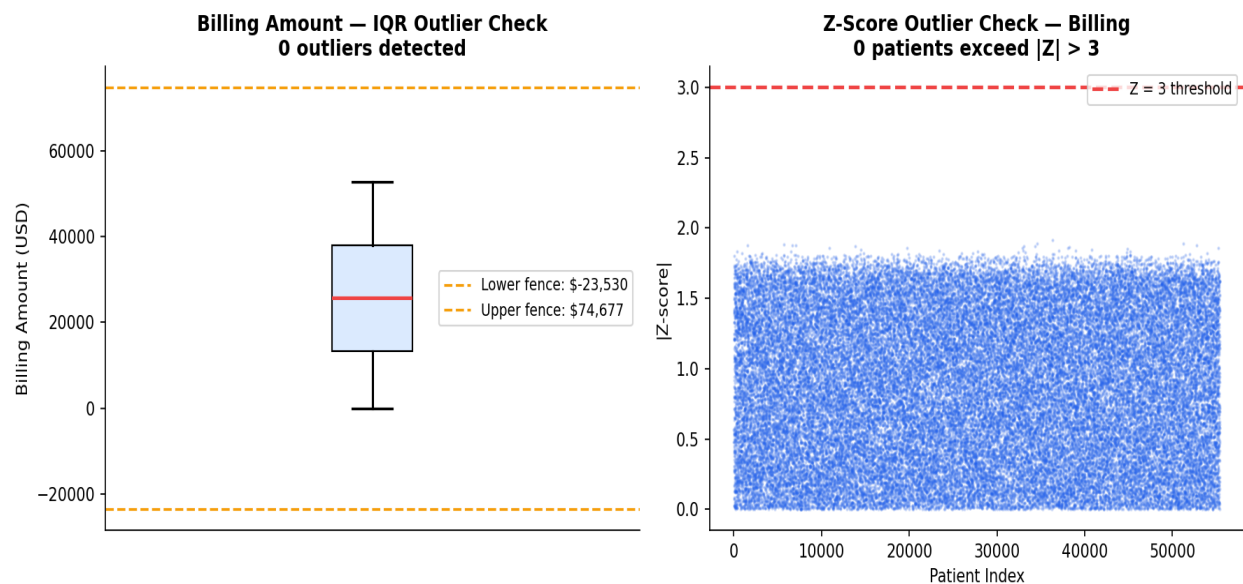


Figure 6 — Outlier detection: IQR fences (left) | Z-score scatter (right)

Statistical Hypothesis Testing

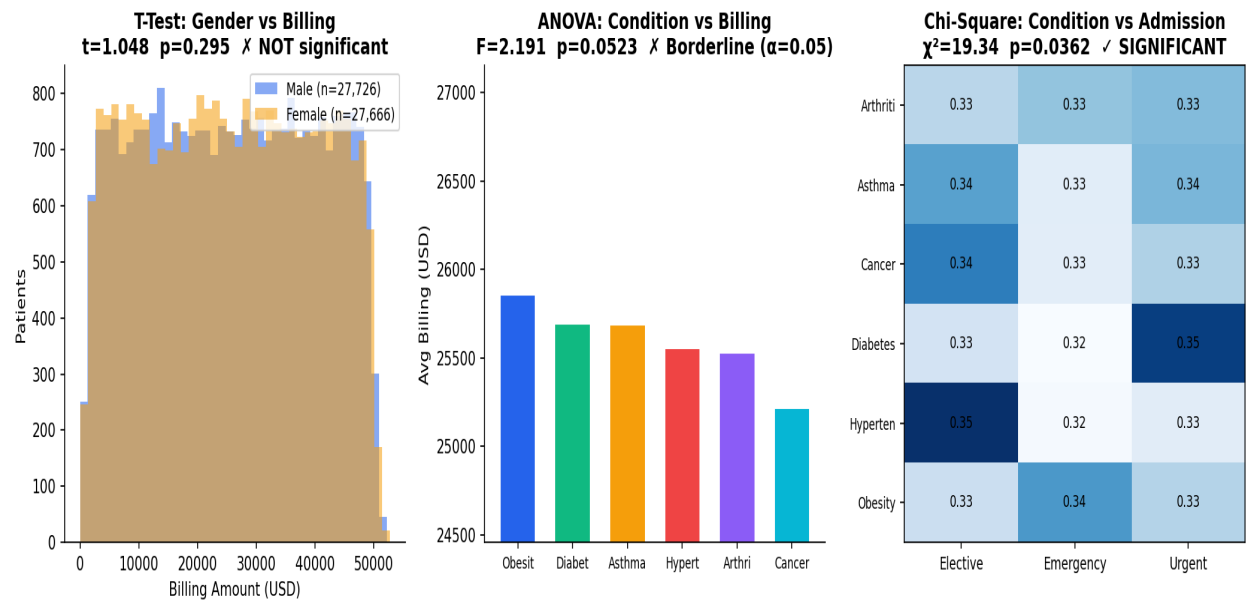


Figure 7 — Statistical tests: T-test (left), ANOVA (centre), Chi-square heatmap (right)

8. Regression Analysis

Objective

Linear regression was applied to answer: which of the available clinical features best predicts billing amount, and how much billing variation can they explain? Five features were encoded and used: Age, Days Hospitalized (LOS), Gender, Medical Condition, and Admission Type.

Feature Encoding

Feature	Encoding	Values
Age	Numeric (no change)	13–89
Days Hospitalized	Numeric (no change)	1–30
Gender	LabelEncoder	Female=0, Male=1
Medical Condition	LabelEncoder	Arthritis=0, Asthma=1, Cancer=2, Diabetes=3, Hypertension=4, Obesity=5
Admission Type	LabelEncoder	Elective=0, Emergency=1, Urgent=2

Model Performance

R ² SCORE	MEAN ABS ERROR	TRAINING SAMPLES	TEST SAMPLES
≈ 0.0002	\$12,361	44,313	11,079
explains ≈0% of variance	48% of avg billing	80% split	20% split

$R^2 \approx 0$ means the model explains essentially none of the billing variation. A Mean Absolute Error of \$12,361 on a mean billing of \$25,590 is a 48% error rate — no better than always predicting the mean. This confirms the statistical test findings: billing is effectively random relative to the clinical features available.

Feature Coefficients

Feature	Coefficient (\$)	Direction	Interpretation
Medical Condition	+\$52.26	Increases billing	Negligible — \$52 on a \$52,764 range
Gender	+\$46.34	Increases billing	Negligible — \$46 difference Male vs Female
Admission Type	-\$52.60	Decreases billing	Negligible — \$53 effect per unit change
Days Hospitalized	-\$6.53	Decreases billing	Counter-intuitive but negligible — within noise
Age	-\$1.30	Decreases billing	Negligible — \$1.30 per additional year of age

Regression Analysis — Linear Model Results

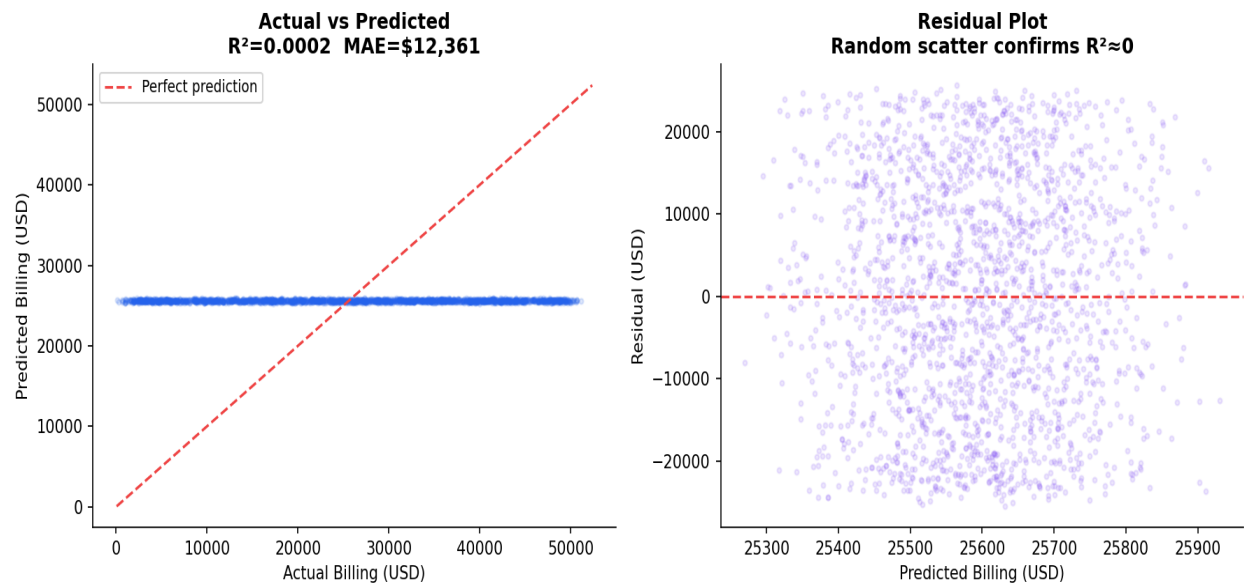


Figure 8 — Actual vs Predicted (left) | Residual plot (right)

9. Recommendations & Conclusion

Priority Actions

Recommended actions for each analytical finding. Ordered by business impact.

Priority	Finding	Recommendation	Channel
Critical	33.6% abnormal test rate (18,627 patients)	System-wide clinical quality audit across all departments. Rate consistent across all 6 conditions confirms no single department is responsible.	Internal clinical review + dashboarding
Critical	Avg LOS = 15.5 days	Introduce structured discharge planning from Day 3. Targeting 13–14 days saves \$1,500–\$2,000 per patient.	Clinical protocol + patient comms
High	Chi-square $p=0.036$ — condition and admission type related	Investigate the admission pathway per condition — statistically detectable relationship warrants operational review.	Clinical + admin review
High	Billing $R^2 \approx 0$ — not predictable	Expand the dataset — add procedure codes, lab test data, and room type to produce a usable billing model.	Data engineering + finance team
High	Summer peak demand	Build a pre-season capacity plan each May ahead of the June–August surge.	Operations + staffing
Medium	ANOVA $p=0.052$ — borderline condition effect	Monitor with a larger time window — borderline result in a 55,500-record dataset is worth tracking as data grows.	Analytics team
Low	108 negative billing records	Investigate source — likely credit adjustments. Implement a billing validation rule to prevent future negatives.	Finance team

Conclusion

55,500 patients were analysed end-to-end — from raw CSV through feature engineering, 13-chart EDA, outlier detection, three statistical tests, and a regression model. Every visual finding was validated before being reported.

The headline finding is counter-intuitive: billing is statistically independent of every clinical variable available (except for a borderline ANOVA signal). This is not a data quality issue — it is a data completeness issue. The variables that actually drive billing (procedures performed, room type, lab tests ordered) are not in this dataset. The Chi-square finding is the standout signal: admission type and condition ARE statistically related ($p=0.036$), a relationship that warrants operational investigation.

Metric	Result
Patients analysed	55,500
Total billing covered	\$1.42B
Abnormal test rate	33.6% (18,627 patients)
Billing outliers (IQR + Z-score)	0 — uniform distribution confirmed
T-test (gender vs billing)	p = 0.295 — NOT significant
ANOVA (condition vs billing)	p = 0.052 — Borderline, NOT significant at $\alpha=0.05$
Chi-square (condition vs adm.)	p = 0.036 — SIGNIFICANT
Regression R ²	≈ 0.0002 — billing not predictable from clinical features
Highest avg billing condition	Obesity (\$25,860)
Peak demand season	Summer (June–August)
Recommended immediate action	System-wide clinical quality audit — 33.6% abnormal rate

Python · Pandas · NumPy	Matplotlib	SciPy · Scikit-learn	MySQL 8.0	ReportLab
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